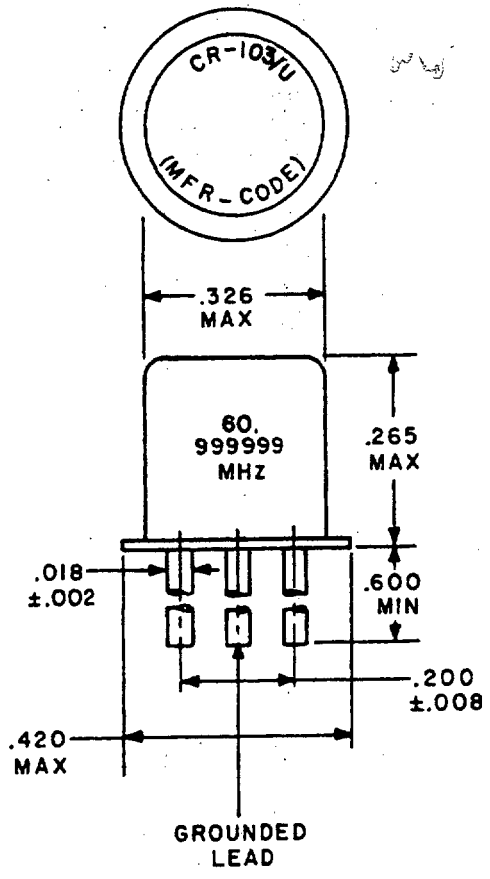
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-103/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 17 to 61 MHz; third overtone; noncontrolled; series.



Inches	mm
.002	0.05
.008	0.20
.018	0.46
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-35/U (MIL-H-10056/21)
Dimensions, marking, and configuration	See figure 1
Frequency range	17 to 61 MHz, inclusive
Frequency tolerance	±25 ppm
Equivalent resistance	50 ohms, maximum
Mode of oscillation	Third overtone
Operating temperature range	-55°C to +105°C, inclusive
Government reference test set	TS-683/TSM
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Drive level, rated	2.0 ±0.4 mW
Capacitance, shunt	7 pF, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent-resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent-resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent-resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±2.5 ppm

1/ Method 204, MIL-STD-202, test condition B.

NOTES

(D) Qualification inspection reinstated as of the date of revision D of this document.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0638)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.